System and Network Administration

Viva Questions and Answers (30 questions)

1. What is a PSU? What is the full form of SMPS?
Ans: PSU converts AC mains supply into low-voltage DC for the PC parts. SMPS means Switched Mode Power Supply.

2. Which connector powers the motherboard?
Ans: The 24-pin ATX connector.

3. What are the PSU wire colours and their voltages?
Ans: Yellow is +12 V, red is +5 V, orange is +3.3 V, black is ground (0 V).

4. What is the use of the green wire in the PSU?
Ans: It is PS_ON. Shorting it to ground switches the PSU on.

5. Why do we wear an anti-static wrist strap?
Ans: To stop static electricity (ESD) from damaging the components.

6. What is a motherboard? What is a chipset?
Ans: The motherboard is the main circuit board that connects all PC parts. The chipset controls data flow between the CPU, RAM and other devices.

7. Why is the CMOS battery needed?
Ans: It keeps the BIOS settings and the clock running when the power is off. It is a CR2032 coin cell.

8. What is the difference between DIMM and SO-DIMM?
Ans: DIMM is the RAM used in desktops. SO-DIMM is smaller and used in laptops.

9. What is the difference between BIOS and UEFI?
Ans: UEFI is the modern replacement for BIOS. It has a graphical menu, supports GPT disks above 2 TB, has Secure Boot and starts faster.

10. What are ACPI and the S3 state?
Ans: ACPI is the standard for power management by the OS. S3 is Sleep: RAM stays powered and everything else is off.

11. What is Wake on LAN?
Ans: A feature that turns on a PC remotely using a special network packet.

12. How do you rotate the screen? What is Extend display?
Ans: Go to Display settings and choose the orientation (Ctrl + Alt + Up arrow restores normal). Extend display spreads the desktop across two monitors as one big workspace.

13. What will you do if the microphone does not work?
Ans: Check the cable, the default input device, the privacy permission and the audio driver.

14. What is encryption? What is symmetric and asymmetric encryption?
Ans: Encryption changes readable data into unreadable cipher text using a key. Symmetric uses one key (AES). Asymmetric uses a public and private key pair (RSA).

15. What is EFS and the cipher command? Why back up the key?
Ans: EFS is the NTFS feature that encrypts files for each user, and cipher is the command to use it. It does not work on Windows Home. If the key is lost, the data cannot be recovered.

16. What are WinRE and Safe Mode?

Ans: WinRE is the Windows Recovery Environment, a set of repair tools used when Windows does not start. Safe Mode loads only the minimum drivers and services to find problems.

17. What is the difference between System Restore and System Reset?

Ans: Restore goes back to an earlier saved state (restore point). Reset reinstalls Windows.

18. What is DiskPart? What is a partition?

Ans: DiskPart is a Windows command-line tool to manage disks. A partition is a logical division of a physical disk.

19. What is the difference between MBR and GPT?

Ans: MBR supports disks up to 2 TB and 4 primary partitions. GPT supports bigger disks and up to 128 partitions.

20. What is a process? What is a PID?

Ans: A process is a program that is running. PID is the unique Process ID given to it.

21. What is the difference between kill and kill -9?

Ans: kill asks the process to stop politely (SIGTERM). kill -9 forces it to stop (SIGKILL).

22. What is an IP address? What is DHCP?

Ans: An IP address uniquely identifies a device on a network. DHCP automatically gives IP addresses to devices.

23. What is DNS? What is a default gateway?

Ans: DNS changes domain names (google.com) into IP addresses. The default gateway is the router that connects our network to other networks.

24. What is a MAC address? What does ping do?

Ans: MAC is the 48-bit physical address of the network card. ping checks whether another device is reachable.

25. What is a VPN? Why does the IP change with a VPN?

Ans: A VPN is a secure encrypted tunnel over the internet. The IP changes because our traffic leaves through the VPN server.

26. What is e-waste? Why is it harmful?

Ans: Old or broken electrical and electronic items. They contain lead, mercury and cadmium that pollute soil and water.

27. What is EPR?

Ans: Extended Producer Responsibility. Manufacturers are responsible for the safe disposal of their products at end of life.

28. What is a UPS? Why is earthing needed in a lab?

Ans: UPS gives backup power during power cuts. Earthing protects people and equipment from electric shock and leakage current.

29. What is a firewall?

Ans: A system that filters network traffic using security rules and blocks unwanted access.

30. What is the 3-2-1 backup rule?

Ans: Keep 3 copies of data, on 2 different types of media, with 1 copy kept off-site.